Combinatorial lower bounds for torus grids and positive word grids

A complete proof of the requested lower bounds

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Abstract

We prove, with the conventions of the supplied problem brief, that

$$\nu_{G_{a,k}}(t) = \frac{t}{2}\,\frac{(a-1)ka}{2} \qquad (a \geq 2,\ k \geq 1,\ 0 \leq t \leq 2/a).$$

We give a uniform, explicitly specified finite construction solving every problem $P(a,k)$. Independently, we prove the factor bound directly for every positive word with one-cycle permutation. This fulfills the alternative expressly permitted in Target B and removes the need for the proposed torus-word-to-torus-grid move equivalence. We also prove the subsequence statement and the requested insertion assembly. Only the facts designated [Lean] or [Written] in the brief are used as preexisting results. No published grid-homology theorem or holomorphic result is an additional dependency.

1 Conventions and a comparison principle

All grid coordinates, rectangles, gradings and canonical classes have exactly the conventions of Section 2 of the brief. In particular, marking coordinates refer to squares, state coordinates to grid points, and the full differential counts empty rectangles regardless of their X-markings. Integers modulo n are compared using representatives in $\{0,\dots,n-1\}$.

Definition 1.1. For a nonzero chain z and $0 \leq t \leq 2$, write

$$L_t^G(z) = \max_{U^e x \in \operatorname{supp} z} \left(\frac{t}{2}A_G(x) - e\right).$$

Let $\mathcal{Z}(O)$ be the set of degree-zero cycles representing the canonical class in the complex with O-markings O. This notation is legitimate: the graded complex, its differential and its polynomial subcomplex depend only on O. The canonical class, being the unique nonzero element of the polynomial image, also depends only on O.

Lemma 1.2 (finite minimum). For every grid of size at least two and $0 \leq t \leq 2$,

$$\nu_G(t) = \min_{z \in \mathcal{Z}(O_G)} L_t^G(z) \in \mathbb{R}.$$

Proof. There is at most one monomial on each state in degree zero, so this chain group is finite-dimensional over the finite field $\mathbb{F}_2$. There are therefore finitely many chains and finitely many canonical representatives. This set is nonempty, and none of its elements is zero because the canonical class is nonzero. By Fact 2.6.3, a chain is in the weighted filtration at level s exactly when each monomial in its support has level at most s. Consequently the infimum in Definition 2.6.4 is the displayed finite minimum. This also proves finiteness for every intermediate link grid used below. □

Lemma 1.3 (comparison with a rescaled parameter). Let G, Q be grids of the same size with the same O-markings. Suppose $b > 0$, $h \in \mathbb{R}$, and

$$A_G(x) \geq h + bA_Q(x) \qquad \text{for every state } x.$$

For every $t \geq 0$ such that $t \leq 2$ and $bt \leq 2$,

$$\nu_G(t) \geq \frac{th}{2} + \nu_Q(bt).$$

Proof. For every integer e and every state x,

$$\lambda_t^G(U^e x) \geq \frac{th}{2} + \lambda_{bt}^Q(U^e x).$$

The representative cycles in the two complexes are exactly the same set $\mathcal{Z}(O)$. For each such cycle, taking the maximum over its support preserves the inequality. Its Q-level is at least $\nu_Q(bt)$ by Lemma 1.2. Taking the minimum of the G-levels proves the claim. This argument compares different parameters directly; it does not invoke a same-parameter filtered-map theorem. □

2 Auxiliary grids with zero threshold

Definition 2.1. For $n \geq 2$ and any permutation O of $\mathbb{Z}_n$, define

$$Q(O): \qquad O_Q(c) = O(c), \qquad X_Q(c) = O(c-1 \bmod n).$$

This is a grid because $c \neq c-1$ and O is injective. Denote by e_c its horizontal row arc, with O in column c and X in column $c+1$. All these arcs advance exactly one column cyclically.

Lemma 2.2 (explicit reduction). Every $Q(O)$ is related by an explicitly determined finite sequence of grid moves to

$$B_2: \qquad O = (1,0), \quad X = (0,1).$$

Proof. For each current size $n \geq 3$, perform the following operations.

First cyclically translate rows to put e_0 in the bottom row. Move that row upwards until the row immediately above it is e_1 or e_{n-1}, stopping at the first such encounter. Every row passed has the form e_j with $2 \leq j \leq n-2$. Its planar horizontal interval $[j, j+1]$ is disjoint from $[0,1]$, including their endpoints. Each swap is therefore an allowed ordinary row commutation. Since both neighboring edges occur somewhere above the initial bottom row, this procedure stops.

If e_1 is above e_0, the adjacent rows, in columns $0,1$, have O at the southwest and northeast squares, X at the southeast square, and a blank northwest square. The other marking of the upper row is its X in column 2. Delete column 1 and the lower row by an O-destabilization, placing the merged O at the northwest square.

If e_{n-1} is above e_0, cyclically translate columns by $+1$. Old columns $n-1, 0, 1$ become columns $0, 1, 2$. The upper row now has O in column 0 and X in column 1, and the lower row has O in column 1 and X in column 2. The southwest square of the block in columns $0, 1$ is blank. Delete column 1 and the upper row by an O-destabilization, placing the merged O at that southwest square.

The blank square in either case is indeed unmarked: its row has its other marking in column 2, distinct from $0, 1$ because $n \geq 3$. Both destabilizations merge the edges $0 \longrightarrow 1$ and $1 \longrightarrow 2$. After deleting column 1, their merged edge again advances one column, and every other edge still advances one column cyclically. Thus the new grid is again $Q(O')$ at size $n-1$.

For explicit relabelling, let r_0, r_1 be the rows of these two edges just before destabilization, and let ρ be the increasing relabelling after deleting r_0. Then

$$O'(0) = \rho(r_1), \qquad O'(c) = \rho(O(c+1)) \quad (1 \leq c \leq n-2).$$

The remaining X-markings obey $X'(c) = O'(c-1)$, including column 0, whose X is in row $\rho(O(n-1)) = O'(n-2)$.

The size decreases once in each iteration. At size two, a cyclic row translation if needed gives B_2. This specifies all moves and terminates after $n-2$ destabilizations. When $n = 3$, there are

no intervening nonneighboring edges, so the same procedure applies without commutations. No commutation of rows sharing an endpoint is used. □

Lemma 2.3 (zero auxiliary threshold). For every $n \geq 2$, every $O \in \mathfrak{S}_n$ and $0 \leq u \leq 2$,

$$\nu_{Q(O)}(u) = 0.$$

Proof. The two states of B_2 have gradings

x	$M_O(x)$	$M_X(x)$	$A(x)$
$(0,1)$	0	-1	0
$(1,0)$	-1	0	-1

as follows by substitution into Definition 2.4.1 of the brief. For each source state there are exactly two empty rectangles. From $(0,1)$, both have O-count zero; from $(1,0)$, both have O-count one. Thus the two terms cancel in each case over $\mathbb{F}_2$. Only $(0,1)$ can carry an integer Laurent exponent in degree zero, and that exponent is zero. It is the canonical anchor by Fact 2.7.3, and its level is zero for all $u \in [0,2]$. Therefore $\nu_{B_2}(u) = 0$. Lemma 2.2 and the already-formalized native move invariance, Fact 2.8.2, give the result. No topological identification of the auxiliary grids with an unknot is required. □

Definition 2.4 (Alexander contribution). For any grid $G = (O, X)$ of size n, set

$$C_G(c,r) = \sum_{j=0}^{c-1} \left(\mathbf{1}_{\{X(j)<r\}} - \mathbf{1}_{\{O(j)<r\}} \right), \qquad K_G = \frac{J(O,O) - J(X,X) - (n-1)}{2}.$$

The sum is empty when $c = 0$.

Lemma 2.5 (Alexander contribution formula). For every state x,

$$A_G(x) = K_G + \sum_{c=0}^{n-1} C_G(c, x(c)).$$

Proof. Subtracting the definitions of the two Maslov gradings gives $A_G(x) = J_{\mathrm{c}}(x,X) - J_{\mathrm{c}}(x,O) + K_G$. For a single state point (c,r), twice the centered-J difference is

$$\sum_{j \geq c} \left(\mathbf{1}_{\{r \leq X(j)\}} - \mathbf{1}_{\{r \leq O(j)\}} \right) + \sum_{j<c} \left(\mathbf{1}_{\{X(j)<r\}} - \mathbf{1}_{\{O(j)<r\}} \right).$$

The first sum equals the second: both marking permutations have exactly $n - r$ values at least r, and complementary inequalities are strict as displayed. Hence the centered-J difference is $C_G(c,r)$. Summing over the state proves the formula, including boundary rows and columns. □

Lemma 2.6 (auxiliary Alexander grading). For every $Q = Q(O)$ and every state x,

$$A_Q(x) = -T_O(x), \qquad T_O(x) = \#\{c : x(c) > O(c-1 \bmod n)\}.$$

Proof. Put $p = O(n-1)$ and $b_c = O(c-1)$. The sum defining C_Q telescopes:

$$C_Q(c,r) = \mathbf{1}_{\{p<r\}} - \mathbf{1}_{\{b_c<r\}}.$$

For $c = 0$ the right side is also zero. Since x is a permutation,

$$\sum_c C_Q(c, x(c)) = (n-1-p) - T_O(x).$$

The word of values of X_Q is obtained from the word of values of O by moving its last value p to the front. Pairs not involving this value are unchanged. The number of increasing pairs involving it changes from p to $n-1-p$. Thus

$$J(X_Q, X_Q) - J(O,O) = n - 1 - 2p, \qquad K_Q = p - (n-1).$$

Lemma 2.5 now gives $A_Q = -T_O$. □

3 The torus-grid lower bound: Target A

Definition 3.1. For $y \in \mathfrak{S}_N$, write

$$F(y) = \sum_{c=0}^{N-1} |y(c) - c|, \qquad D(y) = F(y) - \mathrm{inv}(y), \qquad d(y) = \#\{c : y(c) < c\}.$$

As in the brief, $\theta_s y$ adds s to all values modulo N, and $\bar{x}(c) = -x(c) \bmod N$.

Lemma 3.2 (one value rotation). For all $N \geq 2$ and $y \in \mathfrak{S}_N$,

$$D(\theta_1 y) - D(y) = N - 1 - 2d(y).$$

Proof. Let $p = y^{-1}(N-1)$. All value comparisons not involving column p are unchanged by the rotation. The maximum at p becomes the minimum: its $N-1-p$ inversions with later columns disappear, and its p pairs with earlier columns become inversions. Hence

$$\mathrm{inv}(\theta_1 y) - \mathrm{inv}(y) = 2p - (N-1).$$

For $c \neq p$, the change in $|y(c) - c|$ is -1 when $y(c) < c$ and $+1$ otherwise. Column p is not deficient, and its contribution changes by $p - (N-1-p) = 2p - (N-1)$. Therefore

$$F(\theta_1 y) - F(y) = (N - 1 - 2d(y)) + 2p - (N-1) = 2p - 2d(y).$$

Subtracting proves the claim. This includes $p = 0$ and $p = N-1$; the corresponding collections of inversion pairs are empty. □

Lemma 3.3 (rotation inequality). For every $N \geq 2$, $y \in \mathfrak{S}_N$ and integer $1 \leq a < N$,

$$D(\theta_a y) - D(y) \geq a(N-a) - 2a\, d(y).$$

Proof. For $0 \leq s < N$, exactly s values wrap under addition of s modulo N. A nonwrapping value increases, so it cannot create a new deficiency. Consequently $d(\theta_s y) \leq d(y) + s$, including $s = 0$. Summing Lemma 3.2 for $\theta_s y$, $0 \leq s < a$, gives

$$D(\theta_a y) - D(y) = a(N-1) - 2\sum_{s=0}^{a-1} d(\theta_s y) \geq a(N-1) - 2ad(y) - a(a-1).$$

The last expression is $a(N-a) - 2ad(y)$. All wrapping contributions have been counted explicitly. □

Lemma 3.4 (the particular auxiliary grid). Define

$$B_N : \qquad O(c) = N - 1 - c, \qquad X(c) = -c \bmod N.$$

Then $B_N = Q(O)$, its threshold is zero on $[0, 2]$, and

$$A_{B_N}(x) = -d(\bar{x}) \qquad (x \in \mathfrak{S}_N).$$

Proof. The equality $X(c) = O(c-1)$ is immediate, so Lemmas 2.3 and 2.6 apply. To identify the count in Lemma 2.6, put $y = \bar{x}$ and let p be the column where $y(p) = x(p) = 0$. For $c > 0$ with $c \neq p$, the inequality $x(c) > N - c$ is equivalent to $y(c) < c$. If $p > 0$, column p contributes one to $d(y)$ and zero to $T_O(x)$, while column 0 contributes zero to $d(y)$ and one to $T_O(x)$. If $p = 0$, neither exception contributes. Thus $T_O(x) = d(y)$ in both cases.

For this particular family, the moves can be shortened to exactly $N-2$ destabilizations. In columns $0, 1$ and rows $N-2, N-1$, the markings are O northwest and southeast, X northeast, and a blank southwest square. For $N \geq 3$, the other marking in column 0 is X at row $0 \neq N-2$,

so the blank is genuine. Delete column 1 and row $N-1$, placing O at $(0, N-2)$. Old columns $c \geq 2$ become $c-1$, and the resulting permutations are exactly those of B_{N-1}. $\square$

Theorem 3.5 (M806, including the sharp endpoint). For every $a \geq 2$, $k \geq 1$, put

$$q = ka + 1, \qquad N = a + q, \qquad g = \frac{(a-1)(q-1)}{2}.$$

Then

$$\nu_{G_{a,k}}(t) = \frac{tg}{2} \qquad (0 \leq t \leq 2/a).$$

In particular,

$$\nu_{G_{a,k}}(2/a) = \frac{g}{a} = \frac{(a-1)k}{2}.$$

Proof. For a state x, put $y = \bar{x}$. The written grading formulas 4.2.2 and 4.2.5 of the brief give

$$A_{G_{a,k}}(x) = \frac{D(\theta_a y) - D(y) - (N-1)}{2}.$$

By Lemmas 3.3 and 3.4,

$$A_{G_{a,k}}(x) \geq \frac{a(N-a) - (N-1)}{2} - a\, d(y) = g + aA_{B_N}(x).$$

Here the constant identity is $a(N-a) - (N-1) = aq - a - q + 1 = (a-1)(q-1) = 2g$. The two grids have exactly the same O-markings. Apply Lemma 1.3 with $b = a$, $h = g$ and $Q = B_N$. Because $0 \leq at \leq 2$,

$$\nu_{G_{a,k}}(t) \geq \frac{tg}{2} + \nu_{B_N}(at) = \frac{tg}{2}.$$

The canonical anchor $x^O(c) = -c$ has $M_O = 0$ and $A = g$ by Fact 2.7.3 and Lemma 4.2.5 of the brief. Its level is $tg/2$, proving the reverse inequality. There is no restriction to $k = 1$, and the proof applies unchanged when $a = 2$ or $t = 0$. $\square$

4 Uniform finite certificates and Target C

This section makes the solution of $P(a, k)$ explicit as a finite formula. The construction depends only on N, and the same set works for every required parameter pair of that size.

Definition 4.1 (allowed columns and rectangle matrix). Fix a required size $N = (k+1)a + 1$ and set

$$E_N = \{y \in \mathfrak{S}_N : D(y) \text{ is even and } D(y) \geq 2d(y)\}.$$

Order permutations lexicographically. Over $\mathbb{F}_2$, form the matrix R_N with columns E_N and rows $\{*\} \cup \mathfrak{S}_N$, putting the row $*$ first and ordering the other rows lexicographically. Define

$$R_N(*, y) = \mathbf{1}_{\{y = \mathrm{id}\}}, \qquad R_N(v, y) = \#\,\mathrm{Rect}^\circ(y, v) \bmod 2.$$

Set $b_* = 1$ and $b_v = 0$ for every permutation row.

These entries are completely finite and use no marking restrictions. More explicitly, for $u, v \in \mathbb{Z}_N$ let $\Delta(u, v)$ be the representative of $v - u$ modulo N. For each ordered pair $L \neq R$, include a contribution precisely when $v = y \circ (L\ R)$ and there is no column c satisfying both

$$0 < \Delta(L, c) < \Delta(L, R), \qquad 0 < \Delta(y(L), y(c)) < \Delta(y(L), y(R)).$$

Sum these contributions modulo two. This formula includes rectangles wrapping in either direction, as well as intervals with empty interior.

Lemma 4.2 (consistency of the certificate equations). The system $R_N\eta = b$ is consistent. *Proof.* By Lemma 3.4, $\nu_{B_N}(2) = 0$. The dual certificate theorem, Fact 2.6.5, therefore provides a functional on $C_0(B_N)$ that annihilates boundaries, takes value one on the canonical anchor, and vanishes below level zero. For $y = \bar{x}$ of even defect, Lemma 4.2.2 of the brief and Lemma 3.4 give

$$M_O(x) = -D(y), \qquad \lambda_2^{B_N}(U^{-D(y)/2}\bar{y}) = \frac{D(y)}{2} - d(y).$$

Thus the functional's possible support consists exactly of columns in E_N. Its value on the anchor gives the row $*$ equation.

For an odd-defect permutation v, the degree-one monomial on $\bar{v}$ is $U^{(-D(v)-1)/2}\bar{v}$. Reflection, Fact 2.8.6, puts its outgoing empty rectangles to $\bar{y}$ in bijection with the rectangles from y to v. The Maslov rectangle formula, Fact 2.4.2, makes every corresponding output exponent exactly $-D(y)/2$. Consequently the equation that the functional annihilates this boundary is precisely the row v of $R_N\eta = b$.

For an even-defect v, all entries in that row vanish. Indeed, $F(y)$ is even because the sum of positive displacements equals the absolute sum of negative displacements. Hence $D(y)$ has the parity of $\mathrm{inv}(y)$. A transposition changes that parity, so an empty rectangle from an even-defect y can only end at an odd-defect state. This treats every row of the system. □

Definition 4.3 (an explicit determinant formula). Let r be the largest order of a square minor of R_N with determinant one. Choose the lexicographically first pair of row and column index sets (I, J) of size r with this property, and put $M = R_N[I, J]$. There is at least one such minor since the anchor column has entry one in row $*$. For $1 \le j \le r$, let M_j be obtained from M by replacing column j with b_I. Define the vector η_N by

$$(\eta_N)_{J_j} = \det(M_j), \qquad (\eta_N)_y = 0 \quad (y \notin J), \qquad Y_N = \{y \in E_N : (\eta_N)_y = 1\}.$$

All determinants and entries are in $\mathbb{F}_2$. This is a finite formula indexed by N; Gaussian elimination is an equivalent implementation.

Lemma 4.4 (verification of the formula). The vector η_N satisfies $R_N\eta_N = b$. *Proof.* The columns indexed by J are independent because their restriction to I is invertible. They span all columns: otherwise subtracting from a column the linear combination matching it on I would give a nonzero residual in a row outside I, producing a nonzero minor of order $r+1$. This contradicts the definition of r. By Lemma 4.2, b is in the column span, so it is a linear combination of the columns in J. Restriction to I determines those coefficients uniquely. Cramer's rule, with $\det M = 1$, gives precisely $\det M_j$. Thus the full vector equation holds. □

Theorem 4.5 (all four conditions of $P(a,k)$). For every $a \ge 2$, $k \ge 1$, the set $Y_{(k+1)a+1}$ solves $P(a,k)$. *Proof.* The anchor row of Lemma 4.4 gives $\mathrm{id} \in Y_N$. Membership in E_N gives even defect. For each target v, its row gives

$$\sum_{y \in Y_N} \#\,\mathrm{Rect}^\circ(y, v) = 0 \quad \text{in } \mathbb{F}_2.$$

To spell out the rectangle cases: if v is not obtained from y by a transposition, its contribution is zero. If it is obtained by swapping columns L, R, the only candidates are the two ordered rectangles (L, R) and (R, L). Zero or two empty candidates contribute zero; exactly one contributes one. Their emptiness is tested by the modular-distance formula in Definition 4.1, with no special exclusion for wrapping rectangles or marking contents. The row equation states exactly that the sum of all these remaining ones is even. Even-defect targets receive none, as proved in Lemma 4.2. These are all possible rectangle cancellations.

Finally Lemma 3.3 gives, for $y \in Y_N$,

$$(a-1)D(y) + D(\theta_a y) \ge a(N-a) + a(D(y) - 2d(y)) \ge a(N-a) = aq.$$

These are the four required conditions. □

Corollary 4.6 (functional and mirror cycle). An explicit dual functional is obtained by setting

$$\varphi_{a,k}(U^{-D(\bar{x})/2}x) = \mathbf{1}_{\{\bar{x}\in Y_N\}}$$

on the degree-zero basis and extending linearly. It annihilates boundaries by the preceding rectangle calculation and has value one on the canonical anchor. Its support lies at or above g/a at $t = 2/a$. For any $0 \le t \le 2/a$, put $\sigma = at/2$. On this support,

$$\lambda_t = (1-\sigma)\frac{D(\bar{x})}{2} + \sigma\lambda_{2/a} \ge \frac{tg}{2},$$

using the written nonnegativity of D from Lemma 4.2.4 of the brief. Thus it certifies the entire interval, not only the endpoint. The corresponding mirror cycle is explicitly

$$z_{a,k} = \sum_{y\in Y_N} U^{D(y)/2}y \in C(H_{a,k}).$$

It has degree $1-N$, coefficient one at id and nonnegative exponents. For an empty rectangle $y \longrightarrow v$, the formula $D(v) = D(y) - 1 + 2|O_H \cap r|$ makes the output exponent $(D(v)+1)/2$, independent of the source rectangle. The same parity row therefore cancels its coefficient, proving $dz_{a,k} = 0$.

Corollary 4.7 (Target C). For every $k \ge 1$, $P(2,k)$ is solved by Y_{2k+3}, and

$$\nu_{G_{2,k}}(t) = \frac{tk}{2} \qquad (0 \le t \le 1).$$

The argument uses the already-formalized native Fact 2.8.2, but none of OSSbook Theorem 10.3.1, Proposition 4.6.15 or Chapter 5 as an external theorem. In particular, it introduces none of the book dependencies that Target C asks to remove.

5 A direct proof for positive word grids

We now establish Statement F directly. This also proves the torus-word lower bound required by Target B's expressly permitted alternative.

Definition 5.1 (a preferred admissible word grid). Let $a \ge 2$ and let $w = (s_0, \dots, s_{\ell-1})$ be a positive word in which every generator $0, \dots, a-2$ occurs. Put $n = 3\ell$ and $\delta = (\ell - a + 1)/2$. Construct its arcs as in Definition 2.11.3 of the brief, ordering rows by height and, within each height, by increasing start column. Denote this grid by W.

Lemma 5.2 (tie choices). Every admissible word grid of w is related to W by explicitly specified row commutations. *Proof.* At an integer strand height, write successive incoming and outgoing columns as I_j, O_j. They satisfy $I_j < O_j < I_{j+1}$, since the incoming and outgoing columns within one occurrence differ by one and occurrences are ordered by letter index. The arcs join O_j to I_{j+1} cyclically. The nonwrapping planar intervals are pairwise disjoint. The unique wrapping interval, drawn between its two endpoints in the planar cut, strictly contains all the others. All endpoints are distinct. Thus any two rows at this height are noninterleaving. If there is only one occurrence, no row swap is needed.

At a half-integer height, each interval is $[3h, 3h+2]$ for a different letter index h, so these intervals are pairwise disjoint. Again the corresponding rows are noninterleaving. In each height block, perform adjacent swaps to sort the start columns into increasing order, choosing the first adjacent inversion at each step. Each is an allowed row commutation, and the number of inversions decreases. This reaches W. □

Definition 5.3 (active arcs, including wraparound). For an arc with start column p, end column q and row rank R, declare it active at the cut through grid points in column c as follows: if $p < q$, require $p < c \le q$; if $p > q$, require $c > p$ or $c \le q$. Equivalently, $c \in (p, q]_\circ$. Let

$F_c(r)$ count active arcs with rank less than r, and put $U(r) = F_0(r)$. Thus $U(r)$ counts wrapping arcs with rank less than r. This convention places the cut before the markings in column c, exactly as required by the centered-J grading.

Lemma 5.4 (contribution from active arcs). For the word grid, and indeed for any grid with these directed horizontal arcs,

$$C_W(c, r) = U(r) - F_c(r).$$

Proof. The contribution of one arc to the prefix sum in Definition 2.4 is

$$\left(\mathbf{1}_{\{q<c\}} - \mathbf{1}_{\{p<c\}}\right) \mathbf{1}_{\{R<r\}}.$$

If $p < q$, its first factor is -1 for $p < c \le q$ and zero otherwise. If $p > q$, it is 1 for $q < c \le p$ and zero otherwise; this is 1 minus the active-arc indicator in that case. Thus the factor is always the wrapping indicator minus the active-arc indicator. Sum over all arcs. The inequalities also cover $c = 0$ and equality at either endpoint. □

Lemma 5.5 (active counts and newly started ranks). At every cut there are exactly a active arcs, with distinct heights. Put $b_c = O_W(c-1)$, and let j_c be the number of active arcs strictly below the active arc at row b_c. Then

$$\sum_{c=0}^{n-1} j_c = \sum_{h=0}^{\ell-1} (3s_h + 1).$$

Proof. Immediately before a letter of index s, the active arcs are one at each integer strand height $0, \dots, a-1$. Across that letter's three columns the successive replacements are: the arc at s is replaced by the middle arc at $s + 1/2$; the arc at $s + 1$ is replaced by the outgoing arc at s; the middle arc is replaced by the outgoing arc at $s + 1$. Thus the active count remains a throughout and returns to the integer heights before the next letter. At cut zero, the active arcs are the unique wrapping connector at each integer height, so this argument starts and closes cyclically.

The arc starting in column $c-1$ is active at cut c, and it has row b_c. At cuts $3h+1, 3h+2, 3h+3$, its zero-based height rank is respectively s_h, s_h, s_h+1. The final cut n is cut zero, so these triples account for all cuts exactly once. Summing gives the formula. □

Lemma 5.6 (the two constant counts). For the preferred word grid,

$$\sum_{r=0}^{n-1} U(r) = \sum_{h=0}^{\ell-1} (3s_h + 2), \qquad J(O_W, O_W) - J(X_W, X_W) = 2\ell - a.$$

Proof of the first identity. At each integer height, the unique wrapping arc starts at the last occurrence, so its row is last in its height class. There are no wrapping arcs at half-integer heights. Therefore a row of integer height s has exactly s wrapping rows below it, and a row of height $s+1/2$ has exactly $s+1$ wrapping rows below it. Each letter contributes its three starting arcs at heights $s + 1/2, s, s + 1$, giving $(s + 1) + s + (s + 1) = 3s + 2$.

Proof of the second identity. Count increasing pairs of marking rows in column order. For pairs from two different letter blocks with different heights, the O and X contributions agree: each block has the same height multiset $\{s, s + 1/2, s + 1\}$ for both markings. Within a single block, the O heights are $(s+1/2, s, s+1)$ and the X heights are $(s, s+1, s+1/2)$; each has two increasing pairs. Hence the total difference from unequal heights is zero.

For equal half-integer heights, start and end columns of the middle arcs have the same order, giving difference zero. At an integer height with m occurrences, the row ranks at the O-columns occur in order $0, 1, \dots, m-1$, while the row ranks at the X-columns occur in order $m-1, 0, \dots, m-2$. The latter has $m-1$ fewer increasing pairs. This also gives zero when $m = 1$.

Every strand position occurs, and the total number of such occurrences is 2ℓ. Summing $m-1$ over the a integer heights gives $2\ell - a$. These cases exhaust all pairs. □

Theorem 5.7 (pointwise word-grid comparison). Let $Q = Q(O_W)$. For every state x,

$$A_W(x) \geq \delta + aA_Q(x), \qquad \delta = \frac{\ell - a + 1}{2}.$$

Proof. For each cut c, the active arc at row b_c has exactly j_c active arcs below it. If $r \leq b_c$, then $F_c(r) \leq j_c$. If $r > b_c$, then $F_c(r) \leq a$ because there are a active arcs. Both cases give

$$C_W(c, r) + a\mathbf{1}_{\{r > b_c\}} \geq U(r) - j_c.$$

Apply this with $r = x(c)$ and sum over columns. Since a state uses every row exactly once, Lemmas 2.5, 2.6 and 5.6 give

$$A_W(x) - aA_Q(x) \geq K_W + \sum_{r=0}^{n-1} U(r) - \sum_{c=0}^{n-1} j_c.$$

Here

$$K_W = \frac{2\ell - a - (3\ell - 1)}{2} = \frac{-\ell - a + 1}{2},$$

and the difference of the two sums is ℓ by Lemmas 5.5 and 5.6. The resulting constant is $(-\ell - a + 1)/2 + \ell = (\ell - a + 1)/2 = \delta$. □

Theorem 5.8 (factor bound and Target B's alternative). For every $a \geq 2$ and positive word w containing every generator, every admissible word grid satisfies

$$\nu_{\mathrm{wordGrid}(w)}(t) \geq \frac{t}{2} \frac{\ell(w) - a + 1}{2} \qquad (0 \leq t \leq 2/a).$$

In particular, Statement F holds. Also, for every $a \geq 2$ and $k \geq 1$,

$$\nu_{\mathrm{wordGrid}(c_a^{ka+1})}(t) \geq \frac{tg}{2} \qquad (0 \leq t \leq 2/a).$$

Proof. Apply Lemma 1.3 to Theorem 5.7 with $b = a$ and $h = \delta$. Lemma 2.3 gives $\nu_Q(at) = 0$, so the bound holds for W. Lemma 5.2 and Fact 2.8.2 transfer it to every admissible choice of word grid.

If the word permutation is an a-cycle, every generator occurs: omission of generator r would make every factor preserve the nonempty proper subset $\{0, \ldots, r\}$, so their product could not be an a-cycle. Moreover its sign gives $\ell \equiv a - 1 \pmod 2$, and full occurrence gives $\ell \geq a - 1$. Thus δ is the nonnegative integer in Statement F, proving that statement.

For c_a^q with $q = ka + 1$, its length is $q(a-1)$ and $\delta = (a-1)(q-1)/2 = g$. Substitution proves the torus-word bound. This is precisely the alternative to an explicit move equivalence authorized in Target B: the lower bound has been proved directly for the word grid itself. No assertion of the separate move equivalence M807 is needed or used. □

6 Subsequences and insertion assembly: Targets D and E

Proposition 6.1 (Target D). For $a \geq 2$, a positive word $w = (s_0, \ldots, s_{\ell-1})$ and an integer $q \geq \ell$, the word w is a subsequence of c_a^q. *Proof.* If $\ell = 0$, use the empty embedding. Otherwise define

$$f(i) = i(a-1) + s_i \qquad (0 \leq i < \ell).$$

Then $0 \leq f(i) < (i+1)(a-1) \leq q(a-1)$, and

$$f(i+1) - f(i) = (a-1) + s_{i+1} - s_i \geq (a-1) - (a-2) = 1.$$

Thus f is strictly increasing and takes values in the positions of c_a^q. Position $f(i)$ is letter s_i in the i-th Coxeter block. This is the required list subsequence embedding. The case $q = 0$ necessarily has $\ell = 0$ and was already covered. □

Lemma 6.2 (reversal of an X switch). If $X' = X \circ \tau_c$ with $\tau_c = (c \;\; c+1 \bmod n)$, then $X = X' \circ \tau_c$. *Proof.* Since $n \geq 2$, the two columns are distinct and $\tau_c^2 = \mathrm{id}$. Compose the first equation with τ_c. This includes $c = n-1$, whose partner is column zero, and requires no separate nonwrapping assumption. □

Proposition 6.3 (Target E, with the required inputs explicit). Let $a \geq 2$, let w have one-cycle permutation, and suppose w embeds as a subsequence of c_a^q, where $q = ka + 1$ and $k \geq 1$. Assume the written deletion statement M804. For every $0 \leq t \leq 2/a$,

$$\nu_{\mathrm{wordGrid}(w)}(t) \geq \frac{t\delta(w)}{2}.$$

This is Proposition 3.3.1 of the brief if its M806 and M807 assumptions are used; Theorem 5.8 supplies its needed torus-word bound directly, so M807 can also be omitted.

Proof. Let S be the image of the chosen subsequence embedding in $\{0, \ldots, q(a-1)-1\}$. Write its complement in increasing order as $j_1 < \cdots < j_s$, where $s = q(a-1) - \ell(w)$. Let w_i be the restriction of c_a^q, in increasing position order, to $S \cup \{j_1, \ldots, j_i\}$. Then $w_0 = w$, $w_s = c_a^q$, and each w_{i+1} is obtained by one insertion. Its insertion position is the number of currently retained positions smaller than j_{i+1}. Every w_i contains w, so every generator still occurs.

Apply M804 to the insertion $w_i = u_i v_i$ and $w_{i+1} = u_i \sigma_{r_i} v_i$. It supplies a grid H_i obtained from the larger word grid by one X switch, with H_i move-equivalent to the smaller word grid. Lemma 6.2 reverses that switch. Facts 2.8.4 and 2.8.2 therefore give

$$\nu_{\mathrm{wordGrid}(w_{i+1})}(t) \leq \nu_{\mathrm{wordGrid}(w_i)}(t) + \frac{t}{4}.$$

The intermediate words may close to links, but all these facts hold for their grids. Their sizes are $3\ell(w_i) \geq 3$.

All thresholds here are finite real numbers by Lemma 1.2, since $0 \leq t \leq 2/a \leq 1$ and the canonical classes are nonzero. Hence one may sum and subtract these inequalities in $\mathbb{R}$, and then view the result in the extended reals. Summing, including the empty sum when $s = 0$, gives

$$\nu_{\mathrm{wordGrid}(c_a^q)}(t) \leq \nu_{\mathrm{wordGrid}(w)}(t) + \frac{st}{4}.$$

The required torus-word lower bound is $tg/2$. It follows either from the assumptions M806, M807 and Fact 2.8.2 in the original proposition, or directly from Theorem 5.8. Finally,

$$s = q(a-1) - \ell(w) = (a-1)(q-1) - 2\delta(w) = 2g - 2\delta(w).$$

Subtracting $st/4 = t(g - \delta(w))/2$ gives the result. For an arbitrary word, an admissible k always exists; for example $k = \max\{1, \ell(w)\}$ makes $ka + 1 \geq \ell(w)$, so Proposition 6.1 applies. □

7 Boundary cases, dependencies and formalization interface

Boundary cases. The auxiliary grids are defined for every size at least two, and every reduction stops at two. The first torus case is $(a, k) = (2, 1)$, with $N = 5$, while the rotation argument works for every $1 \leq a < N$. The case $k = 1$ requires no separate argument. At $a = 2$ the interval is $[0, 1]$ and the threshold is $tk/2$. At $t = 0$ both comparison parameters are zero; at $t = 2/a$ the auxiliary parameter is exactly two, where the directly computed auxiliary threshold remains zero. No assertion is made for $t > 2/a$.

The empty word is included in the subsequence proposition. It cannot have a one-cycle permutation on $a \geq 2$ strands, and it is not used to construct a word grid here. Every word used

for the word-grid proof contains all generators, so each integer strand height has a nonempty occurrence list and $n = 3\ell \geq 3$. Single-occurrence height classes and intermediate links were covered explicitly. Reflection is used only in constructing the dual certificate, with the square and state conventions of Fact 2.8.6; the target torus grid is always the reflected grid $G_{a,k}$.

Dependency table. All citations in the following table refer to the supplied brief, not to additional published results.

New result	Permitted inputs from the brief	New published inputs
Lemmas 1.2–1.3	2.5.2, 2.6.3, 2.7.1–2.7.3	None
Lemmas 2.2–2.3	Move definition 2.8.1; 2.7.3, 2.8.2	None
Lemmas 2.5–2.6	Grading definition 2.4.1	None
Lemmas 3.2–3.4	Permutation definitions; Section 2 above	None
Theorem 3.5	2.7.3; written 4.2.2, 4.2.5	None
Lemmas 4.2–4.4	2.4.2, 2.6.5, 2.8.6; written 4.2.2	None
Theorem 4.5	Lemmas 3.3, 4.2–4.4 above	None
Corollaries 4.6–4.7	Written 4.2.3–4.2.5; rectangle grading	None
Lemmas 5.2–5.6; Theorem 5.7	2.11.3; 2.4.1; move definition 2.8.1	None
Theorem 5.8	2.8.2; Sections 1, 2, 5 above	None
Proposition 6.1	Word and subsequence definitions	None
Proposition 6.3	Written M804; 2.8.2, 2.8.4	None

Native move invariance is a genuine dependency. It is used exactly as the already-formalized Fact 2.8.2 permits. Neither the stronger vanishing conjecture $V(a,k)$, nor an identification with holomorphic knot Floer homology, nor the separate move-equivalence statement M807 is assumed. Target B is fulfilled through its allowed direct-word-grid alternative, proved in Theorem 5.8.

Minimal new combinatorial lemmas for formalization. The following finite statements suffice; all their proofs are given above.

1. For $y \in \mathfrak{S}_N$, the exact identity $D(\theta_1 y) = D(y) + N - 1 - 2\#\{c : y(c) < c\}$ and the bound $\#\{c : (\theta_s y)(c) < c\} \leq \#\{c : y(c) < c\} + s$ for $0 \leq s < N$.

2. For every $O \in \mathfrak{S}_n$, the diagram $(O, c \mapsto O(c-1))$ moves to B_2 by the explicit size-decreasing algorithm of Lemma 2.2, and its Alexander grading is $-\#\{c : x(c) > O(c-1)\}$.

3. For grids with equal O, equality of their graded complexes and polynomial subcomplexes identifies their canonical classes. A pointwise Alexander inequality $A_G \geq h + bA_Q$ implies the rescaled threshold inequality of Lemma 1.3.

4. The matrix with entries equal to empty-rectangle cardinalities modulo two and an anchor row has the explicit solution of Definition 4.3, after consistency is obtained from the auxiliary dual certificate.

5. For the preferred admissible word grid, the finite active-arc counts satisfy $C(c,r) = U(r) - F_c(r)$, $\#\text{Active}(c) = a$, $\sum j_c = \sum_h (3s_h + 1)$, $\sum_r U(r) = \sum_h (3s_h + 2)$, and $J(O,O) - J(X,X) = 2\ell - a$.

6. The strictly increasing list-position map $i \mapsto i(a-1) + s_i$ proves the subsequence statement; the adjacent-column transposition is its own inverse; finite minima justify all threshold subtraction in the insertion assembly.

These statements prove Target A for every required pair, give its full interval extension, prove Target C by solving $P(2,k)$, fulfill Target B's stated alternative, and prove Targets D and E. The direct positive-word proof also establishes Statement F without using the insertion route.